

HEALTH CLUB MANAGEMENT SYSTEM (HCMS)

Master Technical Knowledge Base & RAG Domain Document

1. System Overview & Core Objectives

The Health Club Management System (HCMS) is an enterprise 4-tier full-stack application designed to manage health club memberships, personal trainer assignments, workout routines, diet schedules, Razorpay online payments, and AI-driven Q&A; assistance.

Key Goals: 1. Single Active Subscription enforcement. 2. Stateless JWT authentication with public signup support. 3. Automated Razorpay HMAC-SHA256 payment verification. 4. Decoupled microservice architecture.

2. 4-Tier Microservice Architecture

Tier Component	Default Port	Technology Stack	Role & Description
React 18 SPA Frontend	5173	React 18, Vite, Custom Hooks	Renders Admin, Member, and Trainer UI views
Node.js API Gateway	8085	Express, http-proxy-middleware	Central proxy with pathRewrite & CORS headers
Spring Boot Core Backend	8081	Java 21, Spring Security, MySQL	Executes business logic, JWT validation & persistence
Python RAG AI Chatbot	8090	FastAPI, ChromaDB, LangChain	Retrieval-Augmented Generation Q&A microservice

3. Membership Plans & Single Active Subscription Lock

HCMS offers 4 primary membership plan tiers for gym members:

Plan Name	Duration	Duration Days	Price (INR)	Key Features
Monthly Gym	Monthly	30 Days	₹1	Gym floor access, Locker, Basic workout template
Strength Pro	Quarterly	90 Days	₹1	Assigned trainer, Custom workout plan, Tailored diet
Yoga Plus / Cardio Elite	Half-Yearly	180 Days	₹1	Yoga mobility plan, Breathwork guide, Diet chart
Elite Annual / VIP Annual	Annual	365 Days	₹1	All access, Dedicated personal trainer, Weekly updates

Single Active Subscription Lock Rule:

Members can strictly hold only ONE active subscription plan at a time. If a member attempts to purchase a new plan while their current subscription is still active (renewalDate after today and status is not Expired), Spring Boot Backend throws a SubscriptionConflictException: 'You already have an active subscription. You can purchase a new plan after your current subscription expires.'

4. Payments & Razorpay Integration

Online payments are processed through Razorpay SDK using a secure 2-step verification handshake:

1. Order Creation: POST /api/payments/create-order calls RazorpayClient.orders.create({amount, currency: 'INR'}) returning a Razorpay Order ID (e.g. order_12345).
2. Signature Verification: POST /api/payments/verify receives razorpay_order_id, razorpay_payment_id, and razorpay_signature. The backend computes HMAC-SHA256 hash using Utils.verifyPaymentSignature. Upon validation, payment status is set to SUCCESS and subscription is activated.

5. Roles & Access Control (RBAC)

Spring Security 6 enforces Role-Based Access Control (RBAC) with stateless JWT Bearer Tokens:

- ADMIN: Full system access. Executive dashboard (GET /api/admin/dashboard), payment audits (GET /api/payments/admin), member/trainer/plan deletion.
- MEMBER: Self-service profile, subscribe to plans via Razorpay, request personal trainers (POST /api/trainer-requests), view assigned workouts & diet.
- TRAINER: View assigned member clients, accept/reject requests (PUT /api/trainer-requests/{id}/status), update workout JSON (PUT /api/members/{id}/workout-plan), update diet JSON (PUT /api/members/{id}/diet-plan).
- PUBLIC SIGNUP: Permitted without JWT tokens for POST /api/auth/register, POST /api/members, POST /api/trainers, GET /api/plans, and GET /api/trainers.

6. Key REST API Endpoints Summary

HTTP Method	Endpoint Path	Role Permission	Description
POST	/api/auth/register	Public	User registration & profile provisioning
POST	/api/auth/login	Public	Authenticates credentials & issues JWT token
GET	/api/members	ADMIN, MEMBER, TRAINER	Fetches all member profiles
POST	/api/members	Public / ADMIN	Creates new member profile
PUT	/api/members/{id}/workout-plan	ADMIN, TRAINER	Updates member workout schedule JSON
PUT	/api/members/{id}/diet-plan	ADMIN, TRAINER	Updates member diet plan JSON
POST	/api/payments/create-order	ADMIN, MEMBER	Creates Razorpay Order ID
POST	/api/payments/verify	ADMIN, MEMBER	Verifies Razorpay HMAC signature & activates plan
POST	/api/trainer-requests	ADMIN, MEMBER	Submits request for personal trainer
PUT	/api/trainer-requests/{id}/status	ADMIN, TRAINER	Accepts/rejects trainer assignment request
POST	/api/chatbot/chat	Public	Proxies AI Q&A queries to Python RAG Agent